SYE 1010, Introduction to Systems Engineering

Tuesdays and Thursday 4.00PM to 5: 10 PM, College Park Office Bldg 120

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Tuesday-Thursday 02:00 Pm-04:00 Pm and by appointment, 53 College Park Office Bldg

Instructor Introduction/Welcome

Welcome to SYE 1010, I'm truly excited to have you in this class and look forward to the journey we'll take together this semester. My name is Dr. Sara Amar, and I'll be your instructor for this course. Please know that I am here to support you. Your success and well-being are important to me, and I encourage you to reach out at any time; whether you have a question about the material, need clarification, or just want to discuss ideas related to the course. You can contact me via email or during office hours, and I'm always happy to set up a time to talk. Once again, welcome! I'm looking forward to getting to know each of you and to a fantastic semester of learning together.

Intellectual Diversity

Bowling Green State University is committed to fostering a culture of open inquiry and intellectual diversity within the classroom. This course may include topics or discussions that some people find controversial. These conversations are meant to help you understand the course material—not to push any particular opinion. You'll be graded on your understanding of the subject, not on your personal beliefs. Part of our goal is to help you build critical thinking skills so you can form your own informed opinions on complex issues.

Course Overview, Description, Purpose

Introduction to Systems Engineering will provide students with the knowledge of system approach of solving engineering problems. It will cover the definition of systems, components, and interactions within and out of various systems. It will introduce operations research, optimization techniques, and problem-solving techniques using simplex and Linear Programming. Additional topics may include decision analysis techniques for systems, uncertainty analysis, utility functions, multi-attribute utility functions and analysis, influence diagrams, risk preference, analytical hierarchy and node processes in decision-making.

Course Learning Outcomes & Expectations.

This course will introduce the area of systems engineering and industry standard tools to our students- emphasis is on manufacturing and service industries, but the techniques introduced are applicable to all industries.

Student Learning Outcomes

At the completion of this course, students should be able to:
LO1) Interpret the basic concepts in Systems Engineering
LO2) Be able to choose tools and methods of Systems Engineering
LO3) Identify system development life cycle
LO 4) Analyze systems modeling concepts and methodologies
LO 5) Be able to use commercial systems modeling tools

Pre-requisites/Prior Knowledge

None

Course Structure

Intro to Systems Engineering SYE 1010 is a 15-week course conducted in person. While this is an in-person class, learning materials, assignments, and some learning activities will be accessed using BGSU's Learning Management System called Canvas. When we meet for our regularly scheduled class times, you will be expected to participate in in-class activities, engage with the learning materials, and/or complete assignments. Weekly quizzes will be conducted during class. Please use the course schedule, Canvas announcements, and in-class announcements to keep yourself on track.

You should plan to use your assignments and class discussions to gain information and resources for your final project. Supporting documentation can be found in the course materials and should be used throughout your studies. Keeping up with the course pace is important for you to fully engage with the course content and with your fellow learners.

After completing a learning activity or assignment, I will do my best to provide feedback and complete grading within 7 days of the due date. If I am not able to complete feedback/grading within 7 days, I will inform you.

Every week I have dedicated time scheduled to meet with students, which are what I will refer to as office hours. Office hours are a great time for you to ask questions about course content and assignments and discuss feedback. My office hours are from 02 pm to 04 pm on Tuesday and Thursday. Please email me to schedule an appointment during these times. If there are changes to my office hours, I will notify you in class and via a Canvas announcement.

Course Materials

Students will be provided with the lecture materials and additional handouts which are not available in the text and reference books.

Text Book: Systems Engineering and Analysis (4th or 5th Edition)
Prentice Hall International Series in Industrial & Systems Engineering
By Benjamin S. Blanchard, and Wolter J. Fabrycky

Grading Policy & Assessments/Assignments/Learning Activities

Evaluation Criteria

Assessments	Weight
Assignment 1	10%
Assignment 2	10%
Assignment 3	10%
Group Project	10%
Mid Term Exam	30%
Final Exam	30%
Total	100%

Grading Policy

90%-100%	A
80%-89%	B
70%-79%	C
60%-69%	D
0%-59%	F

AI Use Guidelines

You are welcome/expected to use generative AI tools (e.g. ChatGPT, Dall-e, etc.) in this class. You are responsible for the information you submit based on an AI query (for instance, that it does not violate intellectual property laws, or contain misinformation or unethical content). Your use of AI tools must be properly documented and cited in order to stay within university policies on academic honesty.

Key Assignments/Assessments

Assessments	Assignments	LO	Weight
Assignment 1	Canvas Quiz	LO1	10%
Assignment 2	Canvas Quiz	LO1-2	10%
Assignment 3	Canvas Quiz	LO2-3	10%
Group Project	Canvas Assignment-Homework	LO1-5	10%

Mid Term Exam	Canvas Assignment	LO1-2	30%
Final Exam	Canvas Assignment	LO2-3-4-5	30%
Total			100%

Submitting Assignments/Assessments

Most assignments will be submitted using Canvas. There will be a few assignments throughout the semester where you may be asked to submit a physical assignment during class. I have marked how each assignment should be submitted in the Course Agenda section of this syllabus. The assignments you turn in during class will be returned once they have been graded. The assignments you submit through Canvas will be returned to the same location. I provide feedback on your assignments in Canvas. Review this Canvas Community guide for information about how to view my feedback.

If you are unable to gain access to Canvas to turn in your assignment, please send it by email to samar@bgsu.edu using the conventions given under Communication. I will only accept assignments by email during emergency situations.

Academic Honesty Policy/Code of Academic Conduct

Academic honesty enhances and sustains an environment of ethical and principled intellectual pursuit. BGSU has adopted a Code of Academic Conduct to communicate that cheating, plagiarism, and other forms of academic dishonesty are destructive to the mission of the University. You are expected to have read and understand the Academic Code of Conduct, as unawareness of the policy is not a defense. Violations include, but are not limited to, using unauthorized assistance, materials, or study aids (such as generative artificial intelligence), fabrication, plagiarism, and helping another commit an act of academic dishonesty. as unawareness of the policy is not a defense. Violations include, but are not limited to, using unauthorized assistance, materials, or study aids (such as generative artificial intelligence), fabrication, plagiarism, and helping another commit an act of academic dishonesty.

Make-Up and Late Assignment/Exam Policy

All assignments and exams must be submitted by the class meeting time on the date they are due. Late assignments and exams will not be accepted. There is no partial credit given on late assignments; late assignments will result in "0" points.

Withdrawal Policy

According to the BGSU Add/Drop policy, you have seven (7) calendar days to drop this course. After the Add/Drop deadline has passed, you may withdraw from the course. If you formally withdraw from the class before the 12th week of a 15-week course, you will receive a grade of "W". You are responsible for filing a course withdrawal request that notifies me on or before the last day of the 12th week.

Classroom Procedures/Policies

Communication

The best way to reach me is via email. My official BGSU email address is samar@bgsu.edu. When sending an email message, you can expect a response within 24 business hours. If you have not received a response, please try to contact me again, this time via phone.

For urgent matters, I can be reached by phone at 914 733 8048. Please leave a detailed message if I don't answer.

If you would like to send me email, please add the following to the subject line: "<course prefix>: <Student's last name, first name>". Since I get a variety of emails each day, I do not read all emails I receive. By having this heading in the subject line, I will read your email immediately.

Additionally, I can be reached using the Canvas Inbox. This feature will allow us to communicate via Canvas but will also send us each an Outlook email. Again, please use a subject line that includes"<course prefix>: <Student's last name, first name>". Canvas announcements will be posted frequently on the announcement page or communicated via e mail. You should check the announcement page and your e-mail regularly in order to access course related announcements.

Guidelines for Classroom Conduct

In this course, we value an inclusive, respectful, and collaborative learning environment. To support positive and constructive interactions, please adhere to the following guidelines throughout the course:

1. **Respectful Communication:** Treat all participants with kindness and respect, even when discussing complex or controversial topics. Engage thoughtfully in discussions, and ensure that your comments focus on ideas, not individuals. Personal attacks, offensive language, or any form of disrespectful behavior will not be tolerated.
2. **Active Listening:** Actively listen to your peers' perspectives and engage with their ideas in a constructive manner. In online discussions, this means taking the time to carefully read and reflect on others' posts before responding. Acknowledge differing viewpoints and ask questions that encourage further exploration.
3. **Personal Responsibility:** You are responsible for your learning experience, including actively participating in discussions, completing assignments on time, and seeking help when needed. Be proactive in addressing any challenges you face during the course and communicate with your instructor promptly if you need support.
4. **Confidentiality and Sensitivity:** Maintain confidentiality when discussing personal, professional, or sensitive matters that may arise during class interactions. Respect the privacy of your peers and create a safe space for open dialogue by not sharing or discussing what others disclose in the course outside of class.
5. **Creating an Inclusive Space:** We encourage diverse viewpoints and recognize that each student brings a unique perspective to the class. Engage in discussions with an open mind, and consider how your experiences and insights contribute to a broader understanding of the subject matter.
6. **Managing Difficult Conversations:** Disagreements are natural, but it is important to remain respectful and constructive in our responses. When engaging in sensitive topics, be mindful of

your tone and language. If you feel uncomfortable or encounter a situation you cannot address productively, please reach out to the instructor for support.

7. **Expectations for Discussion Participation:** Online learning requires thoughtful and timely participation. Participate in discussions early in the week to allow meaningful interaction with your peers. Make sure to reference course materials in your responses and strive to expand on your peers' ideas to deepen the conversation.

Device Use and Technical Skills

To ensure a successful learning experience in this course, it is essential that you possess certain technical skills and digital information literacy. Specifically, you are expected to be proficient in using a computer or mobile device to access course materials, participate in online discussions, and complete assignments.

Additionally, you should be comfortable navigating the Canvas learning management system (LMS), using word processing software, and conducting online research. For example, you may need to upload assignments in PDF format, participate in virtual group projects using collaboration tools like Office 365, and utilize online databases for research purposes. Clear instructions and resources will be provided to support students in developing these skills as needed.

As research on learning shows, unexpected noises and movement automatically divert and capture people's attention, which means you are affecting everyone's learning experience if your cell phone, tablet, laptop, etc. makes noise or is visually distracting during class.

For this reason, I…

• ask you to turn off your mobile devices and close your laptops during class.

Device Recommendations

To best engage and interact with activities and assignments in Canvas for this class, you will need a computer or device with the following minimum hardware recommendations according to BGSU ITS:

- i5 or i7 Intel® Core™, Apple M series, or equivalent processor
- Minimum 256GB internal storage, 512GB preferred
- 16GB RAM
- Disclaimer: ITS does not recommend the use of Chromebooks because they are not compatible with many BGSU software applications.

Technology Requirements

To successfully complete learning activities and assignments for this class, you will need a computer or device with access to the following technology tools:

- High-speed internet connection
- Word processor, spreadsheet, and presentation tool (you have access to Microsoft Word, Excel, PowerPoint and more by downloading Office 365 on the BGSU ITS page)
- BGSU email
- Canvas Computer Specifications
- Browser requirements in Canvas

Refer to this BGSU ITS page for information about other tools you have access to through BGSU. Laptops and other technology items may be borrowed from the University Library for a designated period of time by presenting your BG1 Card or another photo ID.

Canvas Support and Guides

To help you navigate and utilize Canvas effectively, we have compiled a list of support resources and guides.

- How do I log into Canvas
- Canvas Student Guides
- Canvas Student Video Tutorials

Technical Assistance

If you need technical assistance at any time during the course or to report a problem with Canvas, Zoom, or another technology, you can:

- Refer to the BGSU ITS Knowledge Base
- Start a chat session with a BGSU ITS representative
- Call BGSU ITS to speak with a representative at 419-372-0999
- For students from the Firelands campus, please contact BGSU Firelands Technology Support Services at 419-372-0626 or otss@bgsu.edu, Monday-Friday from 8:00 a.m. to 5 p.m.

Attendance Policy

Your learning depends on participation in class activities. By attending class regularly, you will have access to discussions, collaborative activities, and in-class review sessions. Your attendance will be recorded each week by completing an exit ticket related to the content and learning activities for that day. Attendance is expected unless you are ill or have a valid reason. Contact me in advance via email if you'll be absent. In this course, missing 3 or more classes will result in dropping your grade 1 letter grade. If you have to miss more than 3 classes, please email me as soon as possible to discuss ways to help you participate in classroom activities asynchronously as appropriate.

Participation/Interaction

Our time in class is the opportunity to actively engage with the material we are exploring. Participation during class is crucial because it is an important avenue for learning. I encourage you to be active in every class session. This participation grade serves as a way to credit you with the effort and work you are putting into the class in and out of the classroom. However, I understand that we all have different levels of comfort regarding speaking in class. There will be a variety of ways to participate in class including speaking to the whole group, engaging in small groups activities and discussions, and completing in-class activities such as exit tickets. If you are fully engaged in at least two of those three activities, you will earn full points for a total of 5. Only participating in one of those activities will earn you half of the participation points for that day. If you are not participating at any of those levels you will not earn points. I will communicate your grade to you every week via Canvas. I encourage you to contact me so we can find ways to make participation work for you in this class. Any class missed will not impact your participation grade as long as you are within your allotted absences.

Class Cancellation

If I need to cancel or alter the modality of class, I will inform you at least 24 hours in advance. If I've decided not to cancel class and instead move the class to Zoom, I will ensure you have access to materials and the Zoom link in our Canvas course. For information about how to download Zoom on your device, please visit the BGSU ITS Knowledge Base page on Zoom access. Please see the University Closure information about what to do if the University closes due to inclement weather or other circumstances. Learning Support Services & Resources

Learning Support Services & Resources

Learning Commons [Bowling Green Campus]

The Learning Commons provides free tutoring services to all BGSU students and is located on the 1st floor of Jerome Library. The Learning Commons provides "one-stop-shop" academic support in the areas of Academic Coaching, Supplemental Instruction, Writing Consultations, Math/Stats Tutoring, subject groups, and individual assistance.

- For more information, visit The Learning Commons course-based tutoring page. For support with a course, please contact the Learning Commons using this form.
- For assistance with the writing process, you may submit your paper for review using the inquiry form on the Online Submission page.

The Teaching and Learning Center [Firelands Campus]

The Teaching and Learning Center at Firelands provides free tutoring services to all BGSU Firelands students and is located on the 2nd floor of the North Building. Contact the Writing Lab if you need online help with the writing process.

University Libraries [Bowling Green Campus]

Looking for help with your research papers? Look no further than the University Libraries! Gain access to a wide array of information services and resources to support your classes, including books, e-books, journal articles, databases, special collections, and comfortable study spaces. Need a charger, laptop, or calculator? We've got you covered! The University Libraries is your go-to resource, anytime, anywhere!

- For immediate assistance with your research, take advantage of our Ask Us! service, where our knowledgeable librarians and library staff can assist you in person at the Information Desk, via phone, email, or online chat.
- You can also schedule an Individual Research Appointment with a librarian to receive personalized help with your research assignments, either in person or virtually.

BGSU Firelands Library [Firelands Campus]

The BGSU Firelands Library supports the teaching, learning, and research mission of BGSU by advancing scholarship and creativity through access to collections and user-centered services that connect faculty and students to high-quality information resources. For more information and research support, contact the library at 419-372-0739; firelib@bgsu.edu.

For immediate assistance with your research, take advantage of the Ask Us! service, where our knowledgeable librarians and library staff can assist you in person, via phone, email, or online chat.

Mental Health [Bowling Green Campus]

Your mental health is important to me. If you are currently experiencing a crisis, call (419-372-2081) or visit the Counseling Center website. An on-call counselor is available Monday-Friday from 8am-5pm. The BGSU Counseling Center also provides a number of services such as individual counseling, Counseling Center groups, and other non-emergency services, and pre-recorded workshops. In collaboration with other campus partners, after-hours mental health support is available during the academic year via the BGSU Crisis Response Team.

Mental Health [Firelands Campus]

Your mental health is important to me. If you are currently experiencing distress, call (419-372-2081) An on-call counselor is available Monday-Friday from 8am-5pm. The BGSU Counseling Center also provides a number of services such as individual counseling, Counseling Center groups, and other non-emergency services. You can also access resources at Wellness @ BGSU Firelands.

Falcon Food Pantry

The Falcon Food Pantry is here to support you with nutritious and accessible food. Students, faculty, and staff are welcome to stop by once a week to select a bag of food and hygiene items. The Falcon Food Pantry is located in 109 Central Hall and is open on Mondays and Thursdays from 12:00 PM to 3:00 PM and Wednesdays from 2:30 PM to 5:30 PM. Please bring your BGSU ID card with you.

University Policy Statements

Student Conduct

A community exists on the basis of shared values and principles. At BGSU, students, recognized student organizations, and student groups are expected to uphold and abide by certain standards of conduct. You should familiarize yourself with the Codes, which create a set of expectations for student conduct, ensure a fair process for determining responsibility when student behavior may have deviated from those expectations, and provide appropriate sanctions when a student and/or student organization has violated the Code(s).

- Students who violate faculty expectations as outlined in the syllabus may be subject to conduct action.
- If you have questions, please do not hesitate to contact the Office of the Dean of Students at 419-372-2843 or odos@bgsu.edu.
- View the Student Handbook.
- Learn more about and view the Code of Academic Conduct.
- Learn more about and view the Code of Student Conduct.

Title IX

Bowling Green State University (BGSU) does not discriminate on the basis of sex and prohibits sex discrimination, including sex-based harassment, in any education program or activity that it operates. BGSU's nondiscrimination policy and grievance procedures can be located on the Title IX page. To report information about conduct that may constitute sex discrimination or to make a complaint of sex discrimination under the Title IX policy, please refer to the See It. Hear It. Report It. page. Faculty members are considered "mandatory reporters" and are required to report incidents of sexual misconduct and relationship violence to the Title IX Coordinator.

If you have any questions about Title IX, you can contact BGSU's Title IX Coordinator in person at, 207 Hayes Hall, by email at titleix@bgsu.edu, or by phone at 419-372-8476.

Accessibility Services

Accessibility Services provides equal access and reasonable accommodations to students with disabilities attending BGSU. Students wishing to discuss their eligibility for such accommodations are encouraged to contact the office at phone: 419-372-8495, fax: 419-372-8496, or email: access@bgsu.edu.

Religious Accommodations

In addition to participation/attendance/absence policies already listed, you may have up to three (3) excused absences each academic semester, without penalty, to take time off for reasons of faith or religious or spiritual belief system or to participate in organized activities conducted under the auspices of a religious denomination, church, or other religious or spiritual organization. You are required to notify me in writing of specific dates requested for excused absences and/or alternative accommodation no later than fourteen (14) calendar days after the first day of instruction for this course. After necessary notification, you should consult with me to determine an appropriate alternative opportunity that will be provided to help you complete your academic responsibilities for the course. These requests will remain confidential. The Office of Registration & Records maintains a list of accommodated religious holidays. Please note that this list is not exhaustive of all holidays. For more information about the Student Religious Accommodations policy, you may contact the Dean of Students at odos@bgsu.edu or 419-372-2843.

Freedom of Expression

Bowling Green State University, as a public institution, is committed to freedom of expression and the rights protected by the First Amendment. First Amendment rights have implications inside the classroom as well. Review the Office of the Dean of Students page regarding Freedom of Expression at Bowling Green State University for more information including the policy. To report concerns please visit the See It. Hear It. Report It. page.

Student Veteran-friendly Campus

BGSU educators recognize student veterans' rights when entering and exiting the university system. If you are a student veteran, please communicate with your instructor so reasonable accommodations can be made for absence when drilling or being called to active duty. Refer to the Veterans page for additional information.

University Closure

In most cases, the University will not close due to inclement weather. This means that for winter conditions, the University will generally remain open unless the Sheriff's Department from the county in which your campus is located declares a Level 3 emergency. Once the decision is made to delay, cancel or close, individuals will be notified through AlertBG, email, the BGSU homepage, BGSU social media channels and the news media (see the Emergency Cancellation Delay and Closing Policy and Procedures document for a list of radio and television stations).

Non-Discrimination in Employment and Education

Bowling Green State University is committed to maintaining an employment and educational environment that is free from unlawful discrimination, harassment, and retaliation. The University prohibits discrimination and harassment on the basis of protected class. Retaliation against any person

for asserting their legal rights to be free from discrimination or harassment, for reporting discrimination or harassment, or for participating in activities is strictly prohibited. You may review BGSU's Non-Discrimination in Employment and Education policy for more information. To report concerns, please visit the See It. Hear It. Report It. page.

Remotely Proctored Testing

This course requires remotely proctored testing, that may include quizzes, regular examinations, and/or final examination, during which you may be required to briefly show your surroundings using your web camera (a "room scan"), wherever you choose to take the exam. By choosing to take the exam in your home/residence hall room, you are consenting to a room scan of the area where you take the exam.

If you do not wish to have your home, residence hall room subject to the room scan, you may take the exam from another location where you won't be interrupted, such as the University Libraries or study spaces on campus. If you are enrolled in a distance program, a local library or a testing center at another educational institution may be an option. You will be required to complete the virtual room scan in the alternative location. It is your responsibility to identify an appropriate location to take your exams.

- The room scan will only be visible to your instructor and BGSU faculty or staff with a legitimate need to review the video. Other students enrolled in the class will not be able to see your room scan.
- If you have or require testing accommodation, please contact the Office of Accessibility Services at access@bgsu.edu or 419-372-8495

Course Schedule/Calendar

Week	Module/Lesson	Activities (In-class and/or out of class)	Assignments	Assignment Due Dates
1	Course Overview Class Strategy	Chapters 1	Discussion of outcomes and expectations	08/26, 6:00PM
2	Introduction to Systems Engineering	Chapters 1	Discussion Questions	09/02, 6:00PM
3	Systems Development & Maintenance Process	Chapter 2,6	Discussion Questions	09/09, 6:00PM
	Assignment 1	See Course Materials/ Key assignments		
4	Systems Integration & Design	Chapter 7	Exercises	09/16, 6:00PM

5	Systems Integration & Design	Chapter 8	Exercises	09/23, 6:00PM
	Assignment 2	See Course Materials/ Key assignments		
6	Requirements, the process, kinds of requirements, writing requirements, traceability	Chapter 10	Case Study	09/30, 6:00PM
7	Requirements, the process, kinds of requirements, writing requirements, traceability	Chapter 12	Exercices	10/07, 6:00PM
8	Midterm Exam			10/14, 6:00PM
9-10	Modeling, history, types, usage	Chapter 7	Exercises	10/21, 6:00PM
	Assignment 3	See Course Materials/ Key assignments		
10-11	Functional Diagrams, DFD's, E-R diagrams and generating LDM, PDM, FSM's for modeling behavior	Book #2	Discussion Points	10/28, 6:00 PM
12	Functional Diagrams, DFD's, E-R diagrams and generating LDM, PDM, FSM's for modeling behavior	Book #2	Study Case, Discussion Points Exercises	11/04, 6:00PM
13	Using commercial modeling software (CORE, Rational)	Book #2	Study Case, Discussion Points Exercises	11/11, 6:00PM

	Assignment 4	See Course Materials/ Key assignments		
14	Using commercial modeling software (CORE, Rational)	Book #2	Study Case, Discussion Points Exercises	11/18, 6:00PM
15	Decision making	Book #2	Study Case, Discussion Points Exercises	11/25, 6:00PM
16	Final Exam			12/02, 6:00PM

I reserve the right to update this syllabus as class needs arise. Be assured that I will communicate to you any changes to our schedule, syllabus, or policies quickly and efficiently through Canvas.